

11.7 Summary and notes

11.7.1 Summary

Idea.

1. A linear hypothesis says that the mean vector lies in a subspace $\mathcal{C}(\mathbf{X}_0)$ of the model space. Its F statistic compares the extra sum of squares $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$, per dimension of the test space, with s^2 . Under normal errors it is $F(r - r_0, n - r, \gamma)$, central under the hypothesis, with γ the squared distance from the mean to the reduced model in units of σ^2 (Definition 11.1.1, Theorem 11.1.1, Proposition 11.1.2).
2. Geometrically, F is a function of the angle between the reduced model's residual vector and the test space. The rejection region is a cone, the statistic is invariant under shifts within the reduced model, rescaling and rotations of the test and residual spaces, and in canonical coordinates every hypothesis is a test that a q -dimensional normal mean is zero. The null distribution holds for every spherically symmetric error law (Proposition 11.2.1, Proposition 11.2.2, Proposition 11.2.3, Proposition 11.2.4).
3. A constraint $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ is a hypothesis about the data only if it is estimable and consistent. Nonestimable constraints cannot be tested at all. For a testable hypothesis, $(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \mathbf{\Lambda}]^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$ is the extra sum of squares of the reduced model with offset $\mathbf{X}\mathbf{b}_0$, whatever the rank of $\mathbf{X}$ or $\mathbf{\Lambda}$, and restricted least squares has a closed form through any generalized inverse (Definition 11.3.1, Proposition 11.3.1, Theorem 11.3.2, Proposition 11.3.3).
4. The F test is the likelihood ratio test. The Wald, likelihood ratio and score statistics are nu , $n \log(1 + u)$ and $nu/(1 + u)$ with $u = qF/(n - r)$, so they always satisfy $\text{Wald} > \text{LR} > \text{Score}$. Calibrated by χ^2 , they disagree and have inflated sizes in small samples (Theorem 11.4.1, Proposition 11.4.2).
5. The noncentral F family has monotone likelihood ratio, and every test invariant under the symmetries of the problem is a function of F . Hence the F test is uniformly most powerful among invariant tests (Lemma 11.4.3, Theorem 11.4.4).
6. Power depends only on γ , increases with it, and grows with replication. In the one-way layout the least favourable configuration for a given range puts two means at the extremes and the rest at the midpoint. Observed power is a function of the p-value and adds nothing to it (Theorem 11.5.1, Proposition 11.5.2, Proposition 11.5.3).
7. For one estimable function the F test is the two-sided t test, and the one-sided version has exact size for one-sided hypotheses. A group statistic qF is the largest squared t statistic over all combinations in the group, which explains why group and individual tests can disagree in both directions (Theorem 11.6.1, Proposition 11.6.2).

11.7.2 Notes and sources

Origins. The ratio of mean squares as a test statistic is Fisher's. It grew out of his analysis of variance in the early 1920s, and Fisher (1925) tabulated it on the logarithmic scale, as

$z = \frac{1}{2} \log F$. Snedecor (1934) introduced the letter F in Fisher's honour and published tables of the ratio itself. The exact t distribution of a regression coefficient, with $n - p$ degrees of freedom, is also due to Fisher (1922). The geometry of [Section 11.2](#) follows the coordinate-free tradition described in the notes to [Chapter 6](#). Christensen (2020, chapter 3) develops testing entirely in that language, with test spaces, offsets and nonestimable constraints. Rencher and Schaalje (2008, chapters 8 and 12) and Seber and Lee (2003, chapter 4) give the matrix treatment for full-rank and rank-deficient models, including the canonical elimination of Exercise (C2). Sen and Srivastava (1990, chapter 3) and Agresti (2015, sections 3.1–3.2) emphasize applied examples. Note that Rencher and Schaalje and Christensen define the noncentrality as half of our γ ([Section 4.1](#)).

Testable hypotheses. The insistence that only estimable functions can be tested goes back at least to Searle (1971), whose treatment of “testable hypotheses” in models not of full rank is the standard reference; Proposition 11.3.1 is a minimal version of his argument. Christensen (2020, section 3.3) shows how a nonestimable constraint reduces to its estimable part, which in our treatment is Proposition 9.2.6.

Likelihood ratio, Wald and score. Neyman and Pearson (1933) introduced the most powerful test and the likelihood ratio principle for composite hypotheses. The large-sample χ^2 calibration of the likelihood ratio is due to Wilks (1938), the Wald statistic to Wald (1943), and the score statistic to Rao (1948). The inequality $\text{Wald} \geq \text{LR} \geq \text{score}$ for linear restrictions in normal regression, and the resulting possibility of conflicting verdicts, were pointed out by Berndt and Savin (1977) and discussed further by Breusch (1979). The “ nR^2 ” form of the score test in Exercise (B2) is widely used in econometrics.

Invariance. Lehmann and Romano (2005) develop the theory of invariant tests and prove that the F test is uniformly most powerful invariant; Theorem 11.4.4 follows their approach in outline. Scheffé (1959) treats the power of F tests in the analysis of variance. The robustness of the null distribution to spherically symmetric errors (Proposition 11.2.4) is part of the theory of spherical and elliptical distributions surveyed by Fang, Kotz and Ng (1990).

Power. The noncentral F distribution was tabulated for power calculations by Tang (1938). The least favourable configuration of Proposition 11.5.2 is the basis of the classical sample-size tables for the analysis of variance. Cohen (1988) popularized effect-size based power analysis. Hoenig and Heisey (2001) explain why “observed power” computed from the data is uninformative, the point of Proposition 11.5.3. Wasserstein and Lazar (2016) discuss common misreadings of p-values. Equivalence tests, which can support a claim of “no important difference”, go back to Schuirmann (1987).

Other. The test of equality of two regressions in Exercise (B4) and Exercise (C1) is due to Chow (1960). Longley (1967) assembled his data to test the numerical accuracy of regression programs ([Chapter 10](#)). Monotonicity properties of the χ^2 , F and t distributions in their degrees of freedom, of the kind behind the dilution effect of Exercise (B2), are studied systematically by Ghosh (1973).

11.7.3 References

- Agresti, Alan (2015). *Foundations of Linear and Generalized Linear Models*. Hoboken, NJ: Wiley.
- Berndt, Ernst R. and Savin, N. Eugene (1977). Conflict among Criteria for Testing Hypotheses in the Multivariate Linear Regression Model. *Econometrica* 45(5), 1263–1277.
- Breusch, Trevor S. (1979). Conflict among Criteria for Testing Hypotheses: Extensions

- and Comments. *Econometrica* 47(1), 203–207.
- Chow, Gregory C. (1960). Tests of Equality between Sets of Coefficients in Two Linear Regressions. *Econometrica* 28(3), 591–605.
 - Christensen, Ronald (2020). *Plane Answers to Complex Questions: The Theory of Linear Models*. 5th edition. Cham: Springer.
 - Cohen, Jacob (1988). *Statistical Power Analysis for the Behavioral Sciences*. 2nd edition. Hillsdale, NJ: Lawrence Erlbaum.
 - Fang, Kai-Tai, Kotz, Samuel and Ng, Kai Wang (1990). *Symmetric Multivariate and Related Distributions*. London: Chapman and Hall.
 - Fisher, Ronald A. (1922). The Goodness of Fit of Regression Formulae, and the Distribution of Regression Coefficients. *Journal of the Royal Statistical Society* 85(4), 597–612.
 - Fisher, Ronald A. (1925). *Statistical Methods for Research Workers*. Edinburgh: Oliver and Boyd.
 - Ghosh, B. K. (1973). Some Monotonicity Theorems for χ^2 , F and t Distributions with Applications. *Journal of the Royal Statistical Society. Series B* 35(3), 480–492.
 - Hoenig, John M. and Heisey, Dennis M. (2001). The Abuse of Power: The Pervasive Fallacy of Power Calculations for Data Analysis. *The American Statistician* 55(1), 19–24.
 - Lehmann, E. L. and Romano, Joseph P. (2005). *Testing Statistical Hypotheses*. 3rd edition. New York: Springer.
 - Longley, James W. (1967). An Appraisal of Least Squares Programs for the Electronic Computer from the Point of View of the User. *Journal of the American Statistical Association* 62(319), 819–841.
 - Neyman, Jerzy and Pearson, Egon S. (1933). On the Problem of the Most Efficient Tests of Statistical Hypotheses. *Philosophical Transactions of the Royal Society of London. Series A* 231, 289–337.
 - Rao, C. Radhakrishna (1948). Large Sample Tests of Statistical Hypotheses Concerning Several Parameters with Applications to Problems of Estimation. *Mathematical Proceedings of the Cambridge Philosophical Society* 44(1), 50–57.
 - Rencher, Alvin C. and Schaalje, G. Bruce (2008). *Linear Models in Statistics*. 2nd edition. Hoboken, NJ: Wiley.
 - Scheffé, Henry (1959). *The Analysis of Variance*. New York: Wiley.
 - Schuirmann, Donald J. (1987). A Comparison of the Two One-Sided Tests Procedure and the Power Approach for Assessing the Equivalence of Average Bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics* 15(6), 657–680.
 - Searle, Shayle R. (1971). *Linear Models*. New York: Wiley.
 - Seber, George A. F. and Lee, Alan J. (2003). *Linear Regression Analysis*. 2nd edition. Hoboken, NJ: Wiley.
 - Sen, Ashish and Srivastava, Muni (1990). *Regression Analysis: Theory, Methods, and Applications*. New York: Springer.
 - Snedecor, George W. (1934). *Calculation and Interpretation of Analysis of Variance and Covariance*. Ames, IA: Collegiate Press.
 - Tang, P. C. (1938). The Power Function of the Analysis of Variance Tests with Tables and Illustrations of Their Use. *Statistical Research Memoirs* 2, 126–149.
 - Wald, Abraham (1943). Tests of Statistical Hypotheses Concerning Several Parameters When the Number of Observations Is Large. *Transactions of the American Mathematical Society* 54(3), 426–482.
 - Wasserstein, Ronald L. and Lazar, Nicole A. (2016). The ASA’s Statement on p-Values: Context, Process, and Purpose. *The American Statistician* 70(2), 129–133.
 - Wilks, Samuel S. (1938). The Large-Sample Distribution of the Likelihood Ratio for Testing Composite Hypotheses. *The Annals of Mathematical Statistics* 9(1), 60–62.